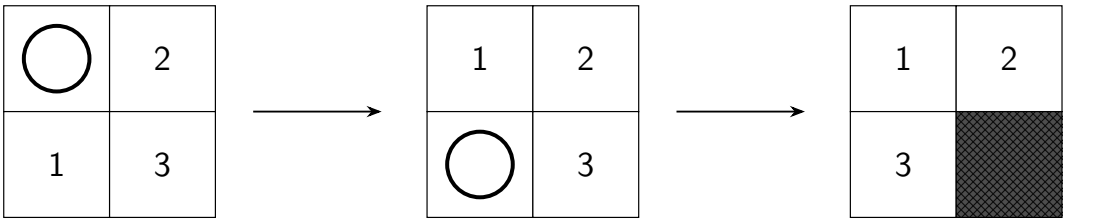
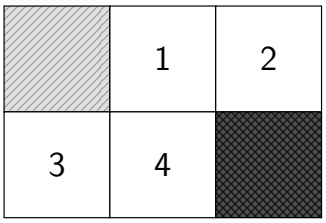
Numbers increase to the right and down. Gray squares are missing; dark squares are outside. Replace a gray corner with the open ring. Slide the smaller numbered neighbor immediately right or below into the ring. If there is only one, use it. Move the ring to the emptied square. Repeat until neither neighbor is numbered; cover that exit with a dark piece.



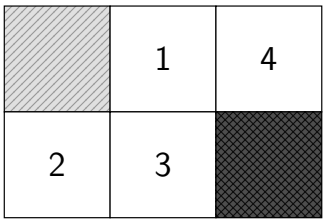
Start | 1 slides up | 3 slides left; cover exit

Problem 1: Build A1, A3 and A5 with the tiles. Close the missing corner in each. Which finish with the same outline?

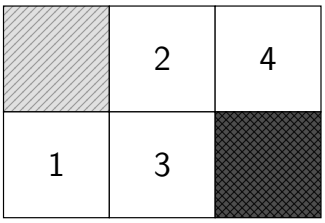
A1



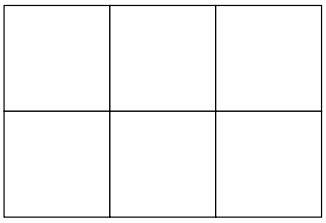
A3



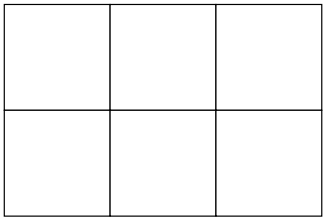
A5



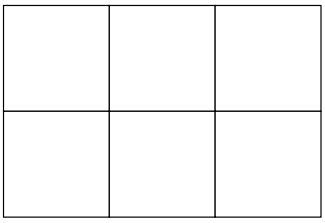
A1



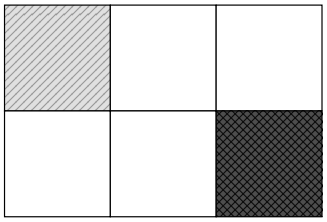
A3



A5



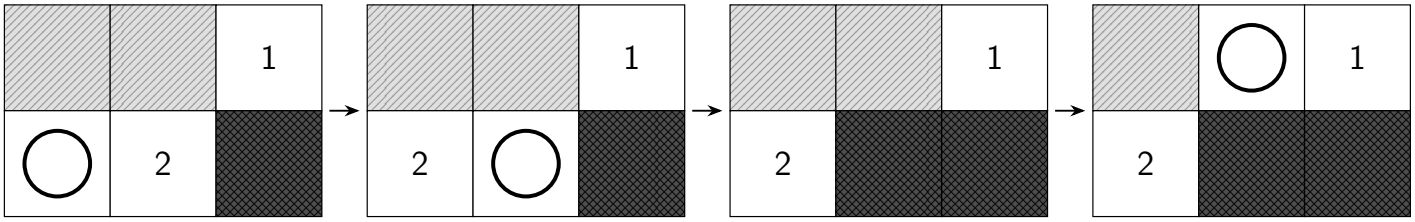
Problem 2: Sort all five A cards by their finished outline. Could an increasing start be missing from this collection? Explain how you know.



Use 1, 2, 3 and 4 once each.

Extra workspace

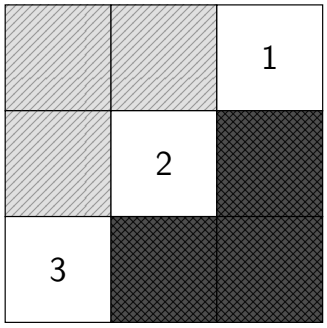
With several gray squares, choose one with no gray square immediately right or below it. Open just that corner. Finish its whole slide and cover its exit before choosing another gray corner. Continue until no gray squares remain.



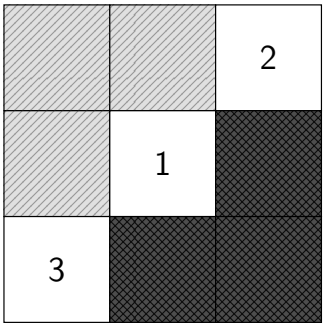
Open lower corner | 2 slides left | Cover the exit | Open next corner

Problem 3: Share the six D starts. Finish each twice: once opening the upper gray corner first, once opening the lower one first. Can that choice change the finished outline or the position of any number?

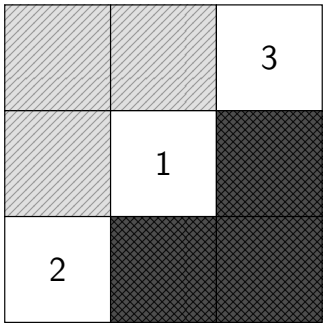
D1



D3

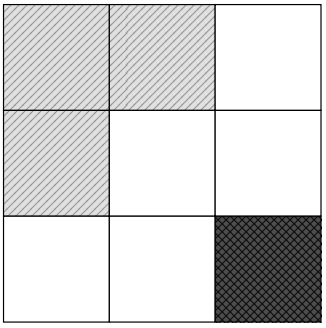


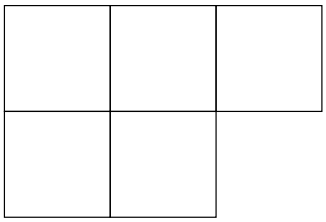
D5

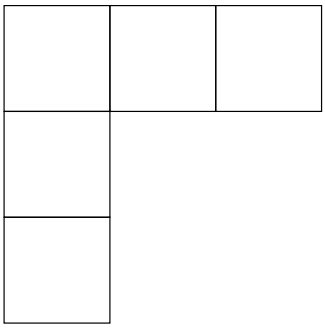


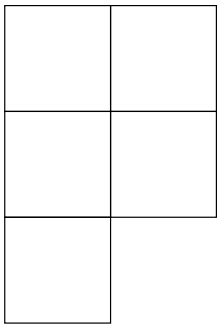
Problem 4: Arrange 1, 2, 3, 4 and 5 on the blank B start so numbers increase right and down. Can you make a start that finishes with each outline below?

B

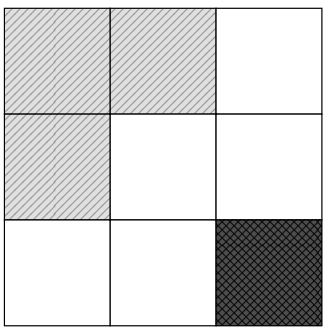


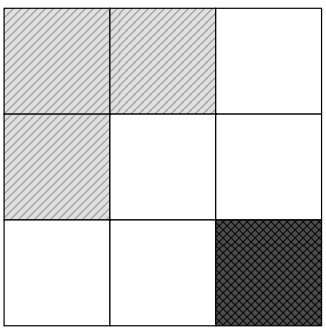


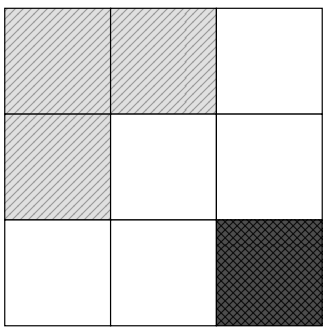




Extra workspace

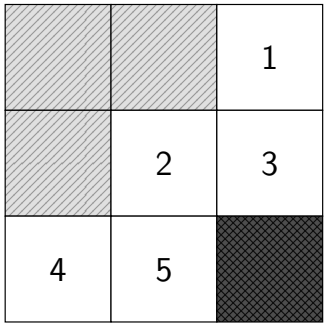




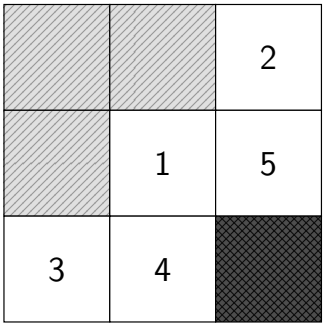


Problem 5: Share out the sixteen B cards. Finish each start with each possible first gray corner. For a single start, can the finished outline or the position of any number differ? Pool the cards and sort them by their finished outlines.

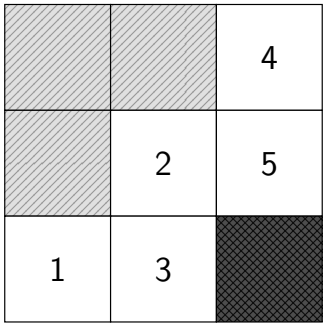
B1



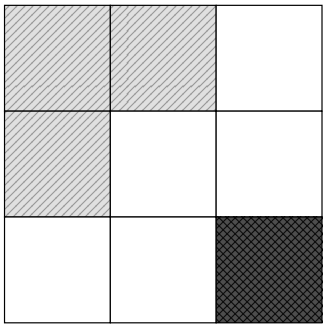
B8



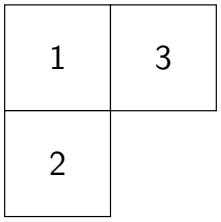
B16

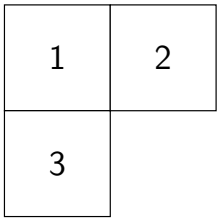


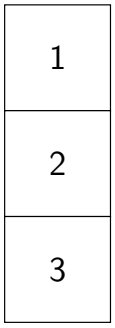
Could an increasing start be missing from the sixteen cards?

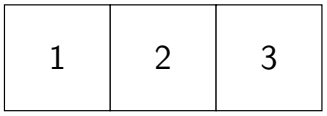


Problem 6: Find every D start that reaches each target. Place its card beside the target. Can a finished arrangement tell you which start was used?



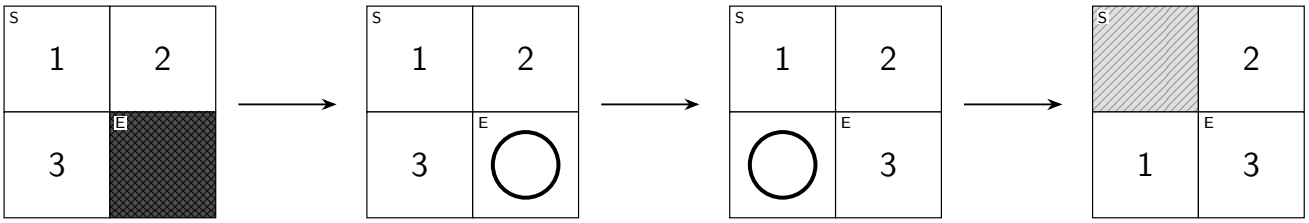




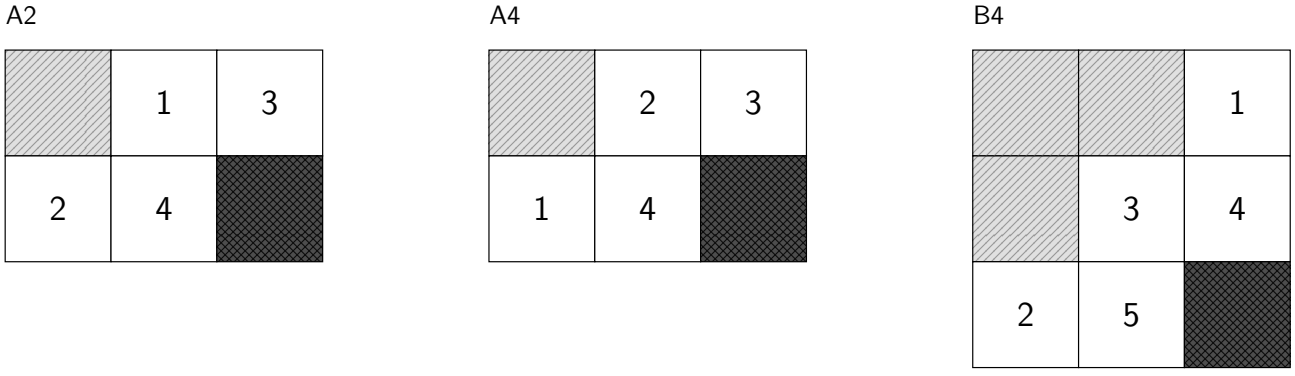


Can two different B starts reach exactly the same finished five-number arrangement?
Use your finished B arrangements from Problem 5.

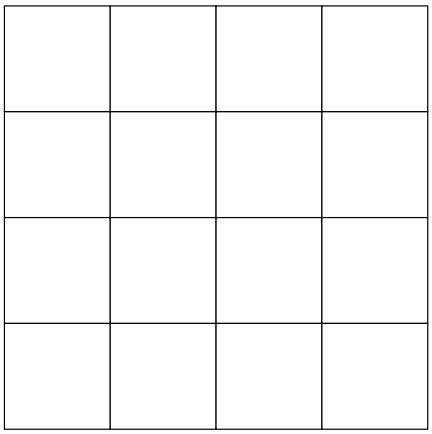
To undo one whole slide, save its starting square S and exit E. Uncover E and put the ring there. Slide in the larger numbered neighbor immediately above or left, or the only one. Move the ring back to the emptied square. Stop when it reaches S; replace the ring with gray.

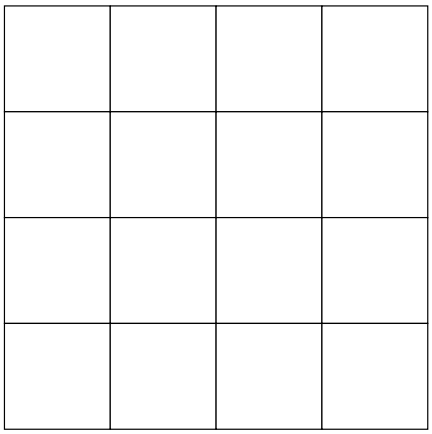


Problem 7: Close the corner in A2 and A4, then undo each slide. Can you recover the start exactly? Try one whole slide from each legal starting corner of B4, then undo it.



Problem 8: Change the rule: slide in the *larger* right or lower neighbor. Find an A start whose finished numbers fail to increase. Does any A start still finish with increasing numbers? What goes wrong?





Problem 9: With the usual smaller-neighbor rule, can a slide ever lose a number, make two numbers fall out of increasing order, or continue forever? Explain what makes each possible or impossible.